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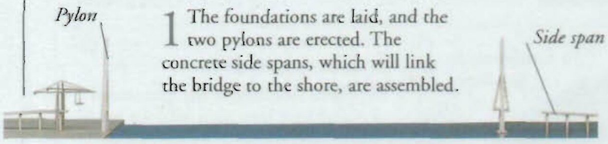
# BRIDGES

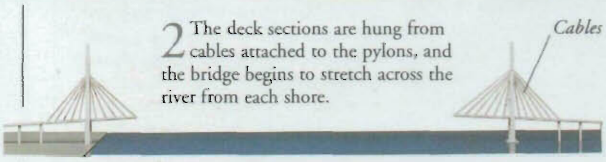
CURVING MAJESTICALLY across rivers and valleys, bridges are some of the most spectacular structures engineers have ever created. They are also some of the most useful, because bridges can speed up journeys by cutting out ferry crossings, long detours, steep hills, and busy junctions. The first bridges were probably tree trunks laid across streams. Wooden beam bridges and stone or brick arches were the main types of bridge from Roman times until the 18th century, when iron became available to engineers. Most modern bridges are made of steel and concrete, making them both strong and flexible.

## Building a bridge

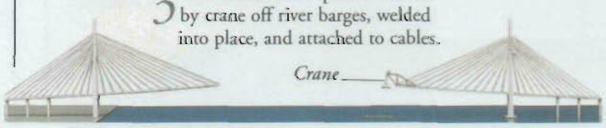
A cable-stay bridge is a type of suspension bridge with a deck hung from slanting cables that are fixed to pylons instead of the ground. Once the pylons are in place, the bridge is built outwards in both directions from each pylon. This ensures that the forces on the pylons balance, so that there is no danger of the pylons collapsing.




1 The foundations are laid, and the two pylons are erected. The concrete side spans, which will link the bridge to the shore, are assembled.




2 The deck sections are hung from cables attached to the pylons, and the bridge begins to stretch across the river from each shore.




3 The central deck spans are lifted by crane off river barges, welded into place, and attached to cables.

4 When the last deck section is in place, the bridge is complete. The cables transfer the weight of the deck to the pylons.


**Model of the Pont de Normandie**

## Types of bridges

On a journey, you may see many different shapes and sizes of bridge, but there are really only a few main types: arch bridges, beam bridges, cantilever bridges, suspension bridges, and cable-stay bridges. The type of bridge used depends on the size of the gap it must span, the landscape, and traffic that will cross it.

**Arch bridge**
The arch is used to build bridges because it is a strong shape that can bear a lot of weight. To bridge a wide gap, several arches of stone or brick are linked together.

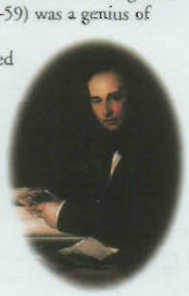
**Beam bridge**
In a beam bridge, the central span (or beam) is supported at both ends. Very long beams are impractical, because they would be liable to collapse under their own weight.

**Cantilever bridge**
A beam fixed at one end and stretching out over a gap is a cantilever. Balanced cantilever bridges have several supports, each with two beams that reach out from either side.

**Suspension bridge**
The deck of a suspension bridge hangs from cables slung over towers and anchored to the ground at each end of the bridge. Such bridges have spans of up to 1 km (0.62 miles).

## Isambard Kingdom Brunel

English engineer Isambard Kingdom Brunel (1806–59) was a genius of bridge design. Brunel designed and built two of the earliest suspension bridges. He also planned and built railways and several huge steamships.



### Aqueducts

Not all bridges carry roads or railway tracks. An aqueduct is a bridge that carries water. The Romans built aqueducts to supply water to the baths and drinking fountains in their cities. More recent aqueducts carry canals over steep-sided valleys in order to keep the canal level. This avoids having to build long flights of locks.

Aqueduct on the River Dee, Wales

### Timeline

**200 BC** Roman engineers build arch bridges of stone or wood, and aqueducts.

**1779** The first bridge made of cast iron is built at Ironbridge, England.

**1883** In the USA, New York's Brooklyn Bridge is the first bridge to be supported by steel suspension cables.

**1930** Switzerland's Salginatobel Bridge is constructed of reinforced concrete (concrete strengthened with steel).

Sydney Harbour Bridge, Australia

**1932** Australia's Sydney Harbour Bridge opens, carrying a road and rail tracks suspended from a huge steel arch.

**1998** The Akashi Kaikyo suspension bridge over Japan's Akashi Strait has the longest main span in the world.

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